KAIZEN MATHS

Lesson starter

Equation of a Circle

Tangents

Name _____

S1

Find the equation of the tangent to the circle at (4, 1):
$(x - 1)^2 + (y + 3)^2 = 25$

S2

Point P is (9, -1). Find the length of a tangent from P to the circle:
$(x + 3)^2 + (y + 1)^2 = 36$

S3

Find the equation of the tangent to the circle at (0, 7):
$(x + 3)^2 + (y - 3)^2 = 25$

S4

Point P is (13, 0). Find the length of a tangent from P to the circle:
$(x - 1)^2 + y^2 = 36$

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KAIZEN MATHS

Exit ticket

Equation of a Circle

Tangents

Name _____

E1

Find the equation of the tangent to the circle at (6, 2):

$(x - 3)^2 + (y + 1)^2 = 18$

E2

Point P is (15, -1). Find the length of a tangent from P to the circle:

$(x - 3)^2 + (y + 1)^2 = 16$

E3

Find the equation of the tangent to the circle at (4, 6):

$x^2 + (y - 3)^2 = 25$

KAIZEN MATHS

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Teacher answers & working

Equation of a Circle · Tangents

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S1

Find the equation of the tangent to the circle at (4, 1):
$(x - 1)^2 + (y + 3)^2 = 25$

Answer:

$y = \frac{-3}{4}x + 4$

Working:

The centre of the circle is (1, -3).

Find the gradient of the radius from (1, -3) to (4, 1).

$m_{radius} = (1 - -3) / (4 - 1) = \frac{4}{3}$.

The tangent is perpendicular to the radius.

$m_{tangent} = \frac{-3}{4}$.

Use y = mx + c with point (4, 1).

The tangent equation is $y = \frac{-3}{4}x + 4$.

S2

Point P is (9, -1). Find the length of a tangent from P to the circle:
$(x + 3)^2 + (y + 1)^2 = 36$

Answer:

$\sqrt{108}$

Working:

The centre of the circle is C = (-3, -1) and the radius is 6.

The radius to the tangent point meets the tangent at 90 degrees.

Use right triangle CPT, where PT is the tangent length.

$CP^2 = (9 - -3)^2 + (-1 - -1)^2 = 144$.

$PT^2 = CP^2 - r^2 = 144 - 36 = 108$.

$PT = \sqrt{108}$.

Teacher answers & working

Equation of a Circle · Tangents

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S3

Find the equation of the tangent to the circle at (0, 7):
$(x + 3)^2 + (y - 3)^2 = 25$

Answer:

$y = \frac{-3}{4}x + 7$

Working:

The centre of the circle is (-3, 3).

Find the gradient of the radius from (-3, 3) to (0, 7).

$m_{radius} = (7 - 3) / (0 - -3) = \frac{4}{3}$.

The tangent is perpendicular to the radius.

$m_{tangent} = \frac{-3}{4}$.

Use y = mx + c with point (0, 7).

The tangent equation is $y = \frac{-3}{4}x + 7$.

S4

Point P is (13, 0). Find the length of a tangent from P to the circle:
$(x - 1)^2 + y^2 = 36$

Answer:

$\sqrt{108}$

Working:

The centre of the circle is C = (1, 0) and the radius is 6.

The radius to the tangent point meets the tangent at 90 degrees.

Use right triangle CPT, where PT is the tangent length.

$CP^2 = (13 - 1)^2 + (0 - 0)^2 = 144$.

$PT^2 = CP^2 - r^2 = 144 - 36 = 108$.

$PT = \sqrt{108}$.

Teacher answers & working

Equation of a Circle · Tangents

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E1

Find the equation of the tangent to the circle at (6, 2):
$(x - 3)^2 + (y + 1)^2 = 18$

Answer:

$y = -x + 8$

Working:

The centre of the circle is (3, -1).

Find the gradient of the radius from (3, -1) to (6, 2).

$m_{radius} = (2 - -1) / (6 - 3) = 1.$

The tangent is perpendicular to the radius.

$m_{tangent} = -1.$

Use y = mx + c with point (6, 2).

The tangent equation is y = -x + 8.

E2

Point P is (15, -1). Find the length of a tangent from P to the circle:
$(x - 3)^2 + (y + 1)^2 = 16$

Answer:

$\sqrt{128}$

Working:

The centre of the circle is C = (3, -1) and the radius is 4.

The radius to the tangent point meets the tangent at 90 degrees.

Use right triangle CPT, where PT is the tangent length.

$CP^2 = (15 - 3)^2 + (-1 - -1)^2 = 144.$

$PT^2 = CP^2 - r^2 = 144 - 16 = 128.$

$PT = \sqrt{128}.$

Teacher answers & working

Equation of a Circle · Tangents

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E3

Find the equation of the tangent to the circle at (4, 6):
$x^2 + (y - 3)^2 = 25$

Answer:

$y = \frac{-4}{3}x + \frac{34}{3}$

Working:

The centre of the circle is (0, 3).

Find the gradient of the radius from (0, 3) to (4, 6).

$m_{radius} = (6 - 3) / (4 - 0) = \frac{3}{4}$.

The tangent is perpendicular to the radius.

$m_{tangent} = \frac{-4}{3}$.

Use y = mx + c with point (4, 6).

The tangent equation is $y = \frac{-4}{3}x + \frac{34}{3}$.